

Online Appendix for: Expected returns in Treasury Bonds

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I. Details on the discounted moving average

I.A. Review of the gain parameters used in the literature

The discounted moving average (DMA), equation (19) in the paper, has been frequently applied in the literature to extract persistent variation in macro data. Table A.1 summarizes representative papers and the value of the gain parameter v that they have used. The star * indicates that the authors calibrate the gain parameter to the data. Given that the gain parameter reflects the strength of per-period updating, many studies refer to economic plausibility arguments, and simply assume some value. While sample periods, types of data and contexts differ across these studies, the gain parameter for quarterly updates lies in a relatively narrow range between 0.90 and 0.98, implying the degree of per-month updating (the frequency which we use in the paper) of 0.985 on average across studies and ranging between 0.974 and 0.995. This is also the range for which we report the sensitivity analysis in Appendix I.D. We show the results in- and out-of-sample for the DMA applied to both core and all-items CPI inflation. In the paper, we use the value $v = 0.987$, which is in line with the 0.95 at the quarterly updating frequency used by Faust and Wright (2013) in the recent survey of inflation forecasting methods.

I.B. Calibration of the gain parameter v to inflation survey data

To provide additional support for the value of the gain parameter we use in our analysis, we calibrate the parameter v to the data. We show that for different assumptions used in calibration, the v parameter is located in a narrow range of values, consistent with these used in the literature summarized in Table A.1. We obtain the v parameter that gives the best fit to inflation surveys and produces the most accurate inflation forecasts. We use nonlinear least squares to fit an AR(1) model in gap form, specified at the monthly frequency as:

$$\pi_{t+\frac{1}{12}} - \tau_t(v, N) = \phi(\pi_t - \tau_t(v, N)) + \varepsilon_{t+\frac{1}{12}}. \quad (1)$$

This model is a simple implementation of the idea that inflation has a persistent long-run mean component (trend inflation, or inflation endpoint), and a mean reverting component. Different, and more complex, versions of this model have been used in the literature to fit inflation dynamics,¹ and the simple gap-AR(1) benchmark has proved to be hard to improve upon out of sample, as recently shown by Faust and Wright (2013). Indeed, in the body of the paper, we show that using τ_t^{CPI} to predict inflation at long horizons delivers the lowest RMSEs among several alternative specifications.

In calibrating the gain parameter we follow Branch and Evans (2006). We measure the mean squared forecast error as:

$$MSE_j = \frac{1}{T_j} \sum_{t=t_j}^{T_j} \left(f_{t+j} - \hat{f}_{t+j}^v \right)^2, \quad (2)$$

where f_{t+j} indicates the forecast that we want to match, and $\hat{f}_{t+j}^v$ is the corresponding forecasts from the gap-AR(1) model (1). Since various inflation forecasts are available at different frequencies and for different measures of inflation, in order not to rely on any specific choice, as f_{t+j} we use the following survey data: (i) four-quarter ahead forecast of GDP deflator from SPF and of CPI inflation from Blue Chip Financial Forecasts, (ii) 10-year ahead inflation forecasts from SPF, (iii) combined average 10-years ahead forecast from Livingston and BCEI from Philly Fed, (iv) one- and six-month ahead survey forecast from Livingston, and (v) one-year ahead realized inflation. The loss function is constructed as the average of MSEs across these measurements in which way we give each of them an equal chance. In the estimation of $\tau_t^{v,N}$, we keep N equal to 120 month as in the body of the paper, and thus v is the only parameter of the discounted moving average that we estimate.

In Figure A.1, we summarize the results for:

- Two measures of inflation: core CPI (rows 1, 2) and all-items CPI (rows 3, 4)
- Two sample periods: 1958:01–2011:12 (rows 1, 3) and 1971:01–2011:12 (rows 2, 4)
- Two sets of results:
 - A Contours of the loss function for different values of v and the mean reversion of the gap ϕ in (1) (panels in column 1): Each plot also reports the v parameter for which the average MSEs are minimized.
 - B Loss function for different v keeping ϕ at a value of 0.95 estimated on a pre-sample, following the “gap-fixed- ρ ” model in Faust and Wright (2013). Accordingly, we estimate the ϕ on the sample period from 1948:01–1958:12 on total CPI, and keep it constant through the rest of the sample. The star marks the minimum value of the MSE.

The calibrated values of v are in line with the literature in Table A.1. Across four cases summarized in Figure A.1, the v parameter (monthly updating) is 0.982, 0.989, 0.985, and 0.985 when we reestimate ϕ , and 0.979, 0.974, 0.984, 0.989 when we keep ϕ fixed at the pre-1958 estimated value. In unreported results, we have excluded some of the measurements, added new measurements (such as short-term surveys with horizon below one year), and consistently obtained the values for v in the range of these reported in Table A.1.

One point that has received relatively little attention in the literature is that of the persistence and the level of long-term inflation expectations across different surveys. In particular, there is evidence of differences between long-term inflation forecasts provided by market participants (the so-called Hoey survey) and by professional forecasters that were quite sizeable in the disinflation period of the 1980s. In the 1980s, market participants had higher projections of inflation than the average of professional forecasters. While the market participants Hoey survey is available only for the period from 1981 to early 1990s, it is the

¹For instance, some examples of the gap-AR model for inflation that has been used in the recent literature are a time-varying parameter VAR of Cogley, Primiceri, and Sargent (2010), an AR(2) of Orphanides and Wei (2012), or AR(12) of Kozicki and Tinsley (2012).

survey that underlies the FRB/US model as discussed Mishkin (2007), and Clark and Nakata (2008).² Interestingly, the discounted moving average with a higher v seems to capture this property of market participants' forecasts quite well especially when applied to the core CPI. The discounted moving average applied to all-items CPI, instead, seems to be closer to professional forecasters survey except in the last part of the post-2000 sample characterized by increased volatility of energy prices (an issue which we discuss in the paper).

I.C. Y-o-y/q-o-q/m-o-m inflation

Figure A.2 shows that it makes little difference whether the discounted moving average is computed using year-over-year, quarter-over-quarter or month-over-month inflation. For year-over-year and quarter-over-quarter we use overlapping data sampled monthly. The gain parameter is $v = 0.987$ as used in the body of the paper. The figure also contains the discounted moving average computed with quarter-over-quarter inflation, sampled at the quarterly frequency with the gain parameter of 0.95. All inflation measures are annualized.

I.D. Sensitivity of predictive results to v and N parameters in constructing τ_t^{CPI}

We analyze the sensitivity of our predictive results to the parameters governing τ_t^{CPI} : the gain parameter v , the length of the moving average window (N). For different combinations of these parameters, we run regressions as in specification (25) in the paper, which is the regression used to construct the cycle factor. In Figure A.3, we plot in-sample and out-of-sample R^2 's varying v between 0.975 and 0.995 and N between 100 and 150 months. We report the results for both core and all-items CPI inflation. While the predictability is stable for both inflation measures and wide range of parameter combinations, it weakens when v approaches unity combined with large window sizes. The deterioration in this region is intuitive as v close to unity over-smoothes the data. As $v \rightarrow 1$, the discounted moving average converges to a simple moving average with the corresponding window. The same pattern is visible out-of-sample for all-items CPI inflation.

I.E. Smoothed past interest rates

In Section 3.2 of the paper, we use moving averages of past interest rates (one- and five-year yield) instead of τ_t^{CPI} in our predictive regressions. The conclusion is that smoothed past interest rates do not have predictive content for bond excess returns. In this Appendix, we additionally present results with the smoothed three-month T-bill rate and consider different smoothing windows. The regression is specified as:

$$\overline{rx}_{t+1} = d_0 + d_1 \bar{y}_t + d_2 y_t^{(1)} + d_3 \tau_t^{y3m} + \varepsilon_{t+1} \quad (3)$$

and τ_t^{y3m} is the smoothed past Tbill rate. We report results where τ_t^{y3m} is obtained as a discounted or simple moving average, considering different smoothing windows. The results are reported in Table A.2 (left panel), and show that τ_t^{y3m} is not a significant predictor of bond excess returns in any of the specifications.

Figure A.4 compares the results from regression (3) with one in which we use smoothed past core CPI inflation (either discounted or simple MA). The figure reports how the in-sample (panel a) and out-of-sample (panel b) predictive results depend on the window size, N with which we smooth either inflation or the Tbill rate. We consider N between 10 and 150 months. The R^2 obtained with the smoothed past T-bill

²The FRB/US series splices data from two surveys of expected long-run inflation—the Hoey survey of financial market participants from 1981 through 1989, and the Survey of Professional Forecasters from 1990 on.

rate are essentially unchanged relative to the yields-only regression, and are not sensitive to the smoothing window used. For the simple moving average of core CPI the results remain relatively stable for windows between 40 and 100 months, and taper off at the extremes. For the DMA there is no similar sensitivity as the window lengthens.³ A very short moving window tilts τ_t^{CPI} to the current realized inflation, which as we show in Section 6.1 of the paper, does not contain additional information about future returns.

One additional test is to introduce jointly the trend inflation *and* the smoothed Tbill rate, in analogy to regression (26) in the paper:

$$\overline{r}x_{t+1} = d_0 + d_1 y_t^{(1)} + d_2 \bar{y}_t + d_3 \tau_t^{CPI} + d_4 \tau_t^{y3m} + \varepsilon_{t+1}. \quad (4)$$

In the paper, we consider the above specification using a smoothed one- or five-year yield, rather than the Tbill rate. The results with the Tbill rate are reported in Table A.2 (right panel), and confirm that it remains statistically insignificant.

We study how the results of regression (4) depend on the strength of smoothing of the Tbill rate. We consider a simple moving average because it requires just one parameter (window size) and thus makes the results easy to summarize. Figure A.5 reports the p-values (based on reverse regressions) testing whether the respective loadings in (4) are zero. We consider three zero restrictions: on the one-year yield $y_t^{(1)}$, on τ_t^{CPI} , and on τ_t^{y3m} . The p-values in Figure A.5 are constructed by varying the moving average window for the Tbill rate between three months and 156 months (13 years). The figure shows that for none of the window sizes the p-value on τ_t^{y3m} falls below 30%, and thus we cannot reject the null that the loading is zero. At the same time, the current short-term yield $y_t^{(1)}$ and the trend-inflation proxy τ_t^{CPI} remain highly statistically significant across all specifications.⁴

Table A.3 presents predictive regressions for individual bond excess returns where the cycle factor is constructed using smoothed past interest rates in place of τ_t^{CPI} . These results correspond to the discussion in Section 6.1 in the paper. The regressions with individual returns confirm the results obtained with average return $\overline{r}x_{t+1}$ that past smoothed interest rates do not contain additional information for predicting returns.

I.F. Random trends

One concern with results reported in Section 3.1 of the paper is that they could arise under entirely random trends that have nothing to do with inflation or the yield curve. We mention this possibility in Section 3.2 of the paper and here provide the details. We set up the null hypothesis that bond returns are predictable with a linear combination of yields. This is the correct null given that the EH has been rejected by numerous studies before and our results alike. By including random trends in our predictive regression, we should recover the predictability that can be attained with yields only, rather than zero R^2 . In Table A.4, we reproduce our regressions using simulated driftless random walks, denoted tr_t^{RW} , in place of τ_t^{CPI} . Under the null, we expect to find that the regression coefficient tr_t^{RW} is not statistically different from zero, and the R^2 equals that from the yields-only regression.

³The effective window size for our parameters ($v = 0.987$ and $N = 120$) is approximately 60 months.

⁴To control for the short rate, in regression (4) we use a one-year yield $y_t^{(1)}$ to be consistent with the approach we have taken in the paper. Our results do not change if we instead used the current (unsmoothed) level of the three-month Tbill rate.

Columns (1)–(7) in Table A.4 summarize the distribution of the regression statistics obtained with 100 000 simulated random trends. The regressions correspond to specifications (24) and (25) in the paper. For ease of reference, columns (8) and (9) provide the respective estimates from the true data. The results indicate that there is less than 0.1% chance to obtain our empirical estimates with random trends. The t-statistics on τ_t^{CPI} and the predictive $\bar{R}^2$ in the data are well above the 99.9th quantile of the simulated distributions (compare columns 7 and 8). The median t-statistics on tr_t^{RW} across simulations is very close to zero (column 3). The $\bar{R}^2$'s are slightly upwardly biased relative to the yields-only regression in the data, but the median bias does not exceed 2% in the cycles specification and 5% in the yields-plus-random trend specification (compare columns 3 and 9). Overall, we conclude that it is very unlikely that our results are spuriously driven by a random trend.

II. Forecasting short rate out of sample

In the paper, we argue that τ_t^{CPI} and $c_t^{(1)}$ do not predict bond excess returns, but they capture the EH term in yields, i.e. forecast the future short rate.

In this section, we forecast the short rate out of sample over annual horizons from one year to five years ahead comparing alternative models. We follow the literature that starts with Fama and Bliss (1987) and use the one-year yield as a proxy for the short rate. The models we consider are:

1. Model based on τ_t^{CPI} formed as in the body of the paper and the corresponding cycles
2. Model based on the discounted moving average of the one-year yield τ_t^{y1} with the same parameters as τ_t^{CPI}
3. Random walk applied directly to the one-year yield
4. Model based on the principal components (PCs) of the yield curve, as in Duffee (2009)

In specifications 1 and 2, the three factors are obtained as follows: (i) τ_t^X is either smoothed core inflation or smoothed one-year yield, $X = \{CPI, y1\}$, (ii) $c_t^{(1)}(\tau_t^X)$ is obtained as a residual from regressing one-year yield on τ_t^X , and (iii) $\widehat{cf}_t(\tau_t^X)$ is obtained as a residual from regression of $\bar{c}_t(\tau^X)$ on $c_t^{(1)}(\tau^X)$,⁵ where $\bar{c}_t(\tau^X)$ is the average cycle across maturities. We explicitly write cycles as functions of τ^X to make clear that they represent the component in yields orthogonal to a particular variable τ_t^X .

In specification 4, the factors are obtained as the first three PCs of the yield curve: level, slope and curvature.

For forecasting purposes, we assume that the most persistent factor, i.e. τ_t in specifications 1–2 and the level factor in specification 4, follows a random walk so that its today's value is the best forecast for its future value. The dynamics of the other two factors, i.e. $c_t^{(1)}(\tau_t^X)$ and $\widehat{cf}_t(\tau_t^X)$ in specifications 1–2, and slope and curvature in specification 4, are modeled as a VAR. The mean reversion matrix in the three-factor VAR has the form:

$$K = \begin{pmatrix} 1 & 0 & 0 \\ 0 & k_{22} & k_{23} \\ 0 & k_{32} & k_{33} \end{pmatrix}, \quad (5)$$

which follows Duffee (2009). Using a three factor (three PCs) model of the yield curve, Duffee (2009) shows that such a constrained specification outperforms other models used in the literature in terms of forecasting interest rates, and is close to par with the random walk.

⁵Section IV of this Appendix compares different ways of constructing the cycle factor $\widehat{cf}_t$. The approach of orthogonalizing the average cycle with respect to the one-period cycle is easy to implement out of sample as it does not require information about future returns. We note that this approach to constructing the cycle factor exploits the result discussed in the paper that the risk premium is orthogonal to the EH term.

Finally, apart from considering three factor models, we also study their more restrictive two-factor versions. In this case, we use as factors τ_t^X and $c_t^{(1)}(\tau_t^X)$ (expectations hypothesis terms) in specifications 1–2 and level and slope in specification 4. Again, the first factor follows a random walk, and the second has a mean reverting AR(1) dynamics, and factors are assumed to be uncorrelated with each other.

Yield loadings on factors are obtained from a linear regression, i.e. we do not impose no-arbitrage restrictions. This approach follows the recent literature (Duffee, 2009; Joslin, Le, and Singleton, 2012) which shows the irrelevance of no arbitrage restrictions for forecasting. Using yield loadings and the above factor dynamics, we form forecasts of the short rate across horizons. The entire procedure (clearly, also construction of factors) is repeated recursively using monthly data up to time t to build forecasts for h months ahead, where h is between 12 and 60 months.

We use three different starting points for the out-of-sample exercise, 1978:01, 1985:01 and 1995:01; each sample period begins in 1971:11. The out-of-sample forecasts are formed using an expanding window.

Table A.5 reports the RMSEs in basis points for the specification 1 above (column 1), as well as its RMSEs relative to the other specifications 2–4 (columns 2–4). The number below one suggests that short rate forecasts based on our decomposition using τ_t^{CPI} are more precise than the ones obtained with the alternative models.

The results show that this is the case for most forecast horizons and alternative models. In general, forecasts that are based on τ_t^{CPI} to identify the EH terms outperform alternatives by a significant margin, including those that use the moving average of the short rate itself (specification 2 with τ_t^{y1}). The gap in the RMSEs becomes most visible at longer forecast horizons. This holds true for two- and three-factor models. Clearly, as the horizon lengthens the statistical power of the exercise deteriorates, but we report the long-horizon forecasts for completeness. Comparing column (1) in Panels A and B, we observe that the three-factor specification does not consistently improve upon the predictability of the two-factor model that includes only the expectations hypothesis terms τ_t^{CPI} and $c_t^{(1)}$.

III. Robustness of predictability results to different bond data sets

III.A. Summary statistics of bond excess returns

Table A.6 contains summary statistics for bond excess returns with a one-year holding period. The returns are computed from the zero-coupon yield curve bootstrapped from the CMT yields.

III.B. Comparison of excess returns from different data sets

Realized bond excess returns are commonly computed using zero-coupon bonds. Since the computation of returns can be sensitive to the interpolation method, we compare returns obtained from CMTs to those from the GSW and FB data. Table A.7 presents the regressions of one-year holding period CMT excess returns on their GSW and FB counterparts with matching maturities. Figure A.6 additionally graphs selected maturities. Excess returns line up very closely across alternative data sets. The R^2 's from regressions of CMT excess returns on GSW and FB consistently exceed 99%, except for the ten-year bond for which the R^2 drops to 98% due to one data point in the early part of the sample (1975). Beta coefficients are not economically different from one.

III.C. Bond return predictability in different data sets

III.C.1. CRSP bond portfolios and predictability at different return horizons

One may be worried that the predictive results depend on the use of splined (artificial) zero-coupon bond returns and on the choice of the annual forecast horizon. To address these concerns, we use actual bond returns for different maturity segments available from the CRSP Fama bond portfolio files for the sample period 1972–2011. The data consists of the monthly equal-weighted average of unadjusted holding period returns for each bond in the portfolio. From monthly returns, we compute returns in excess of the Tbill rate over different investment horizons. Table A.8 contains the predictability results for one-, three-, six-, nine- and twelve-month excess returns. We find that $\widehat{cf}_t$ is a strongly significant predictor of monthly returns at all maturities, explaining between 3% and 5% of their variance. The predictability increases with the return horizon. For the twelve-month holding period, the results broadly agree with those reported in Table 4 of the paper based on splined zero-coupon annual bond returns.

III.C.2. Different ways of computing zero-coupon yields

A related concern is the sensitivity of our forecasting factor to the way we construct the zero-coupon yield curve from CMT yields. To show that our results are robust to this choice, we perform the predictive exercise on the Fama and Bliss (FB) and Gürkaynak, Sack, and Wright (2006, GSW) data sets. These data sets differ not only in the method of splining the zero curve, but also in the choice of the underlying yields. For instance, CMT yields are based on the on-the-run securities, while GSW yields are off the run. We focus on the range of maturities from one to five years, as dictated by the FB data. We additionally consider two samples, 1971–2006 and 1971–2011, recognizing that the dislocations in the Treasury market during the recent financial crisis 2007/09 could affect the construction of zero-coupon yields.

The results are reported in Table A.9 of this Appendix. The forecasting ability of cycles is confirmed across all data sets. We continue to find that two cycles have the predictive power that is comparable to multiple cycles, and therefore this conclusion is not sensitive to the measurement error that the splining of zero-coupon yields induces. The yields-only approach leads to a weaker predictability, diminishing the $\bar{R}^2$'s by a half. The gap between cycles and other predictors becomes larger when we include the crisis years, during which the performance of all regressors weakens, with forward rates explaining about 17% of the variation in $\overline{rx}_{t+1}$. We also note that using the all-items or core inflation to construct τ_t^{CPI} leads to very similar results in the pre-crisis sample, but the discrepancy increases with the inclusion of the crisis years. This is consistent with the fact that all-items inflation in that period reflected unusually high volatility of energy prices, which did not pass through onto the yield curve or onto longer-term inflation expectations, as discussed in the paper. Overall, however, with either core or all-items inflation, the evidence of predictability is significantly stronger and more stable across data sets using cycles than with yields only.

III.C.3. Return horizon used in constructing the forecasting factor

To obtain the single forecasting factor, it is common to project an annual holding period excess return averaged across maturities, $\overline{rx}_{t+1}$, on a set of regressors (e.g., Cochrane and Piazzesi, 2005; Ludvigson and Ng, 2009). We follow the same approach in constructing $\widehat{cf}_t$ in the paper, even though in Section IV of this Appendix we discuss alternative ways to obtain the factor that avoid the use of future returns. Since the choice of the return holding period is arbitrary, other holding periods should be able to pick up the same risk premium component in yields, i.e. we could equally estimate the regression (24) in the paper using a one-month excess return rather than the one-year excess return on the LHS. We verify that our cycle factor is not sensitive to this choice by comparing factors obtained from projections of one-month and one-year holding period excess returns. Across different dataset (CMT, GSW and FB), we find correlations above 0.99 between the “one-month” and “one-year” $\widehat{cf}_t$. The corresponding correlations are 0.87, 0.71, and 0.85 between the “one-month” and “one-year” versions of the Cochrane-Piazzesi (CP) factor.

III.C.4. Number of yields used in constructing the forecasting factor

A related question is how sensitive the cycle factor $\widehat{cf}$ is to the number of yields used on the RHS in its construction. To investigate this sensitivity, we construct $\widehat{cf}_t$ using yields with maturities from one through five years from each of the three data sets (CMT, GSW, FB), as well as seven yields in the CMT data set (maturities up to 20 years) and GSW data set (maturities up to 15 years). We perform an analogous exercise for the CP factor. The results indicate that the CP factor is more sensitive to the choice of the data set and maturities than the $\widehat{cf}_t$. The differences become stronger in real time, where due to the small noise in the data one can obtain linear combinations of forwards whose correlation is below 0.5 across specifications. In the full sample, correlation between various versions of the CP factor ($\widehat{cf}_t$ factor) ranges from 0.66 to 0.93 (0.89 to 1.00) in levels and from 0.29 to 0.74 (0.78 to 0.95) in monthly changes. For the CP factor ($\widehat{cf}_t$) constructed in real time, the correlations range from 0.44 to 0.73 (0.89 to 0.99) in levels and from 0.01 to 0.56 (0.83 to 0.95) in monthly changes. This confirms that the CP factor can be more sensitive to small noise in the data than the cycle factor.

IV. Alternative ways to construct the cycle factor, $\widehat{cf}$

This appendix discusses alternative approaches to constructing the cycle factor introduced in Section 3.3 of the paper. In all specifications we define cycles relative to τ_t^{CPI} constructed as in the body of the paper.

IV.A. Without exploiting information about future returns

The baseline construction of the cycle factor in equation (27) of the paper uses $\overline{rx}_{t+1}$ on the left-hand side. We now use only time- t variables to obtain $\widehat{cf}$, i.e. unlike the CP factor, this way of obtaining the return forecasting factor does not involve future returns. We can estimate the return forecasting factor as the residual from the regression of $\bar{c}_t$ onto $c_t^{(1)}$:

$$\bar{c}_t = \gamma_1 c_t^{(1)} + \bar{\varepsilon}_{t+1}, \quad (6)$$

$$\widehat{cf}_t^{CS} = \bar{c}_t - \hat{\gamma}_1 c_t^{(1)}. \quad (7)$$

The superscript “CS” indicates that we provide only cross sectional information to construct the forecasting factor. Notice that this approach relies on the result in the paper that $c_t^{(1)}$ does not predict bond excess returns and therefore reflects the variation in yields that is orthogonal to the risk premium.

IV.B. One-step NLS estimation

We can also form the cycle factor as a single linear combination of c_t 's:

$$\widehat{cf}_t^{NLS} = \lambda' \mathbf{c}_t, \quad (8)$$

and estimate the restricted system:

$$\mathbf{rx}_{t+1} = A \begin{pmatrix} 1 \\ \lambda' \mathbf{c}_t \end{pmatrix} + \varepsilon_{t+1}, \quad (9)$$

where $\mathbf{rx}_{t+1}$ is a $(m-1) \times 1$ vector of individual returns with maturities from two to $m = 20$ years, $\mathbf{rx}_{t+1} = (rx_{t+1}^{(2)}, rx_{t+1}^{(3)}, \dots, rx_{t+1}^{(m)})'$, $\mathbf{c}_t$ is a vector of cycles, and A is a matrix parameters:

$$A = \begin{pmatrix} \alpha_0^{(2)} & \alpha_1^{(2)} \\ \alpha_0^{(3)} & \alpha_1^{(3)} \\ \vdots & \vdots \\ \alpha_0^{(m)} & \alpha_1^{(m)} \end{pmatrix}. \quad (10)$$

We perform non-linear least squares (NLS) estimation, by minimizing the sum of squared errors:

$$(\hat{A}, \hat{\lambda}) = \min_{A, \lambda} \sum_{t=1}^T \left(\mathbf{r}\mathbf{x}_{t+1} - A \begin{pmatrix} 1 \\ \lambda' \mathbf{c}_t \end{pmatrix} \right)' \left(\mathbf{r}\mathbf{x}_{t+1} - A \begin{pmatrix} 1 \\ \lambda' \mathbf{c}_t \end{pmatrix} \right). \quad (11)$$

For identification, we set $\alpha_1^{(7)} = 1$. This choice is without loss of generality. The loss function (11) is minimized iteratively until its values are not changing between subsequent iterations. In application, being interested in the dynamics of the cycle factor $\widehat{cf}_t^{NLS}$, we additionally standardize excess returns cycles prior to estimation.

IV.C. Common factor by eigenvalue decomposition

Alternatively, in constructing the cycle factor we can exploit the regression of an individual excess return on $c_t^{(1)}$ and the cycle of the corresponding maturity:

$$rx_{t+1}^{(n)} = \alpha_0^{(n)} + \alpha_1^{(n)} c_t^{(1)} + \alpha_2^{(n)} c_t^{(n)} + \varepsilon_{t+1}^{(n)}. \quad (12)$$

We form a vector $\mathbf{erx}_t$ of expected excess returns obtained from this model:

$$\mathbf{erx}_t = E_t \left(rx_{t+1}^{(2)}, rx_{t+1}^{(3)}, \dots, rx_{t+1}^{(m)} \right)'. \quad (13)$$

The single factor is obtained as the first principal component of the covariance matrix of $\mathbf{erx}_t$:

$$\widehat{cf}_t^{PC} = U'_{(:,1)} \mathbf{erx}_t, \quad (14)$$

where $Cov(\mathbf{erx}_t) = U L U'$, and $U_{(:,1)}$ denotes the eigenvector associated with the largest eigenvalue in L . Using returns from two to 20 years, the first principal component explains 93% of common variation in $\mathbf{erx}_t$.

IV.D. Comparing the results

We compare the cycle factor obtained with the different procedures. To distinguish between approaches, we use the notation: $\widehat{cf}_t^{CS}$ for the construction without involving future returns, $\widehat{cf}_t^{NLS}$ for the one-step NLS estimation, $\widehat{cf}_t^{PC}$ for the factor obtained with the eigenvalue decomposition of expected returns, and $\widehat{cf}_t$ for the simple approach introduced in the body of the paper in Section 3.3.

Table A.10 presents the correlations among the four measures. Clearly, while the methods differ, they all identify virtually the same dynamics of the cycle factor. The correlation between the constructed factors exceeds 98%.

V. Details on survey data used in the robustness analysis

V.A. Inflation survey data

In Section 6.1 of the paper, we use various inflation surveys that are available at different frequencies and over different sample period. The various measures are summarized below:

- *Long-horizon inflation expectations from surveys:* (i) combined median ten-year CPI inflation expectations from the Livingston survey and from the Blue Chip Economic Indicators (BCEI) survey, spliced by the Philadelphia Fed; (ii) mean CPI inflation forecast 5–10 years ahead from the BCEI survey (prior to 1984, the variable was the GNP deflator). Both series are forecasts of all-items CPI and are available semiannually from the end of 1979.
- *Short-horizon inflation expectations from surveys:* (i) median forecast of GDP deflator one-year ahead from the Survey of Professional Forecasters (SPF), available quarterly starting from 1967:Q4; (ii) median four-quarters-ahead inflation forecast from the Blue Chip Financial Forecasts (BCFF), available monthly starting from 1984:07.

V.B. Interest rate survey data

We use a range of available interest rate surveys to compare the survey-implied expected returns with the return forecasting factor $\widehat{cf}_t$ that we construct. We use the following survey data:

- BCEI survey: available semiannually in March and October each year over the sample period 1984–2010, contains average forecast of the three-month T-bill over the next ten years. Using this forecast, we construct the risk premium for the ten-year yield, labeled “BCEI 10Y.”
- BCFF survey: available monthly starting from 1988:01, contains four-quarter ahead forecasts of interest rates with seven maturities: 3-month, 6-month, one, two, five, ten and 30-year. The forecasters predict CMT yields, i.e. par coupon yields. Therefore, we first construct the corresponding zero-coupon yield curve of the forecasts, which we then use to compute expected returns as $E_t^s(rx_{t+1}^{(n)}) = -(n-1)E_t^s(y_{t+1}^{(n-1)}) + ny_t^{(n)} - y_t^{(1)}$. With the exception of the two-year excess return, this last step involves cross-sectionally interpolated survey forecast data; e.g. to obtain an expected return on a five-year bond, we need to construct an expected zero coupon yield for a four-year bond.
- SPF survey: available quarterly starting from 1981, contains a one- and two-year ahead forecasts of the three-month T-bill, from which we obtain the survey-implied risk premium for the two-year bond, labeled “SPF 2Y.”

We correlate the survey-implied risk premia from the above sources with the $\widehat{cf}^{RT}$ factor. Since surveys are real time, the forecasting factor is also constructed in real time (hence the superscript *RT*) using two different zero yield data sets: CMT and Fama-Bliss.

Table A.11 displays correlations of various survey-based measures of expected returns with $\widehat{cf}^{RT}$.⁶ All correlations are positive and significant. Correlations are higher for the two-year expected return, and lower for the 10-year expected return. The correlation among survey-based measures ranges from 0.36 (for ten-year maturity) to 0.75 (for two-year maturity). The correlation of survey-based measures with $\widehat{cf}^{RT}$ ranges from 0.34 (for ten-year maturity) to 0.76 (for two-year maturity).

⁶Because of the data availability, we measure these correlations across different maturities and across different definitions of risk premia in order to give as broad as possible range of results. The underlying assumption is that the risk premium variation is dominated by a single factor.

VI. Out-of-sample tests

Below we describe the implementation of the bootstrap procedure to obtain the critical values for the ENC-NEW test. The test statistic for maturity n is given by:

$$\text{ENC-NEW}^{(n)} = (T - h + 1) \frac{\sum_{t=1}^T \left(u_{t+12}^{2,(n)} - u_{t+12}^{(n)} \varepsilon_{t+12}^{(n)} \right)}{\sum_{t=1}^T \varepsilon_{t+12}^{2,(n)}}, \quad (15)$$

where T is the number of observations in the sample, $\varepsilon_t^{(n)}$ and $u_t^{(n)}$ denote the prediction error from the unrestricted and restricted model, respectively, and h measures the forecast horizon, in our case $h = 12$ months. Note that the time step in (15) is expressed in months.

Our implementation of bootstrap follows Clark and McCracken (2005) and Goyal and Welch (2008). To describe the dynamics of yields and to obtain shocks to the state variables generating them, we assume that the yield curve is described by four principal components following a VAR(1). Persistent component τ_t is assumed to follow an AR(12) process. We account for the overlap in bond excess returns by implementing an MA(12) structure of errors in the predictive regression. Imposing the null of predictability by the linear combination of forward rates, we estimate the predictive regression, the VAR(1) for yield factors and VAR(12) for τ_t by OLS on the full sample. We store the estimated parameters and use the residuals as shocks to state variables for the resampling. We sample with replacement from residuals and apply the estimated model parameters to construct the bootstrapped yield curve, the persistent component and bond excess returns. To start each series, we pick a random date and take the corresponding number of previous observations to obtain the initial bootstrap observation. In our case, the maximum lag equals 12, hence we effectively sample from $T - 12$ observations. We construct 1000 bootstrapped series, run the out-of-sample prediction exercise and compute the ENC-NEW statistic for each of the constructed series. We repeat this scheme for different maturities. The critical value is the 95-th percentile of the bootstrapped ENC-NEW statistics.

The out-of-sample R^2 proposed by Campbell and Thompson (2008) is defined as:

$$R_{\text{OOS}}^{2,(n)} = 1 - \frac{\sum_{t=1}^{T-12} \left(rx_{t+12}^{(n)} - \widehat{rx}_{cyc,t+12}^{(n)} \right)^2}{\sum_{t=1}^{T-12} \left(rx_{t+12}^{(n)} - \overline{rx}_{t+12}^{(n)} \right)^2}, \quad (16)$$

where the time step t and sample size T is expressed in months. $\widehat{rx}_{cyc,t+12}^{(n)}$ is the forecast of annual excess return based on time t cycles, where the parameters of the predictive model are estimated using cycles up to time $t - 12$ and returns realized up to time t . $\overline{rx}_{t+12}^{(n)}$ is the return forecast using historical average excess return estimated through time t .

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Table A.1: Overview of gain parameters v used in the literature

The table provides an overview of the constant gain parameters used in the literature. * indicates studies that calibrate the parameter to the data.

Authors	Data	Data frequency	CG parameter	Implied CG for monthly data
Branch-Evans* (2006)	SPF surveys; GDP deflator, GDP growth	quarterly	0.938; 0.953; 0.9655	0.984; 0.988; 0.991
Faust-Wright (2011)	inflation	quarterly	0.95	0.987;
Hemert-Koijen-van Nieuwerburgh (2009)	interest rates	monthly	0.95–0.99; 0.97	0.97;
Kozicki-Tinsley (2001b)	inflation	monthly	0.985	0.985;
Kozicki-Tinsley (2012)	inflation surveys	quarterly	0.96	0.99;
Malmendier-Nagel* (2012)	Michigan survey; GDP deflator; cross section and time series	quarterly	0.98; 0.97	0.995; 0.992
Orphanides-Wei (2012)	GDP growth, inflation, 3-month nominal interest rate	quarterly	0.95	0.987;
Orphanides-Williams* (2005)	SPF; inflation, unemployment and FFR; grid search	quarterly	0.99–0.96; 0.98	0.995;
Piazzesi-Schneider (2011)	inflation surveys; interest rate surveys	quarterly	0.9; 0.95	0.974; 0.987

Table A.2: Predictive regressions with trends based on smoothed past interest rates

The left-hand variable is $\overline{rx}_{t+1}$. Columns present results for different ways of smoothing the past three-month T-bill rate (τ_t^{yld}). DMA denotes the discounted moving average with the same parameters as for τ_t^{CPI} in the body of the paper. MA₁₂, MA₂₄, MA₆₀ denote a simple moving average over the window of 12, 24 and 60 months, respectively. Reverse regression t-statistics are in parentheses.

	Smoothed 3mth Tbill rate				Smoothed 3mth Tbill rate + τ_t^{CPI}			
	DMA	MA ₁₂	MA ₂₄	MA ₆₀	DMA	MA ₁₂	MA ₂₄	MA ₆₀
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$yld^{(1)}$	-0.49	-0.44	-0.42	-0.54	-0.58	-0.69	-0.64	-0.55
	(-2.99)	(-2.32)	(-2.44)	(-3.48)	(-3.51)	(-3.39)	(-3.75)	(-3.47)
$\bar{y}_t$	0.74	0.53	0.59	0.87	1.38	1.44	1.41	1.34
	(2.59)	(2.39)	(2.36)	(2.84)	(4.36)	(5.04)	(4.92)	(4.25)
τ_t^{y3m}	-0.19	0.02	-0.06	-0.23	0.13	0.12	0.12	0.15
	(-1.00)	(0.18)	(-0.36)	(-1.18)	(0.79)	(0.78)	(1.08)	(0.70)
τ_t^{CPI}	—	—	—	—	-1.10	-1.04	-1.07	-1.13
	—	—	—	—	(-4.72)	(-4.58)	(-4.50)	(-4.26)
$\bar{R}^2$	0.21	0.18	0.18	0.22	0.54	0.54	0.54	0.54

Table A.3: Predictive regressions with trends based on smoothed past interest rates

This table reports predictive regressions of individual bond excess returns on the cycle factor (rows (1)–(3)), where the cycle factor is constructed according to equation (27) in the paper but using smoothed past interest rates rather than τ_t^{CPI} . Rows (4) “pval Wald” show p-values for the test that the coefficient on a given trend measure is zero, in the regression specification (24) in the paper. All t-statistics and Wald tests are based on reverse regressions. For ease of comparison, both left- and right-hand-side variables are standardized. The sample period is 1971:11–2010:12.

	$rx^{(2)}$	$rx^{(5)}$	$rx^{(7)}$	$rx^{(10)}$	$rx^{(15)}$	$rx^{(20)}$
A. 60-month simple mov.avg. of 3-month Tbill rate, $y_t^{(3m)}$						
(1) $\widehat{cf}(\text{MA60}, y_t^{(3m)})$	0.27	0.39	0.43	0.48	0.50	0.48
(2)	(1.27)	(2.61)	(3.14)	(3.61)	(4.03)	(4.14)
(3) $\bar{R}^2$	0.07	0.15	0.19	0.23	0.25	0.23
(4) pval Wald (MA60, $y_t^{(3m)}$)	0.58	0.31	0.26	0.20	0.19	0.23
B. Discounted mov.avg. of 3-month Tbill rate, $y_t^{(3m)}$						
(1) $\widehat{cf}(\text{DMA}, y_t^{(3m)})$	0.27	0.39	0.43	0.46	0.48	0.47
(2)	(1.17)	(2.34)	(2.74)	(3.07)	(3.35)	(3.43)
(3) $\bar{R}^2$	0.07	0.15	0.18	0.21	0.23	0.22
(4) pval Wald (DMA, $y_t^{(3m)}$)	0.71	0.38	0.33	0.29	0.28	0.32
C. 60-month simple mov.avg. of 5-year yield, $y_t^{(5y)}$						
(1) $\widehat{cf}(\text{MA60}, y_t^{(5)})$	0.30	0.41	0.44	0.47	0.48	0.46
(2)	(1.41)	(2.58)	(3.01)	(3.36)	(3.66)	(3.79)
(3) $\bar{R}^2$	0.09	0.16	0.19	0.22	0.23	0.21
(4) pval Wald (MA60, $y_t^{(5)}$)	0.46	0.34	0.34	0.33	0.37	0.44
D. Discounted mov.avg. of 5-year yield, $y_t^{(5y)}$						
(1) $\widehat{cf}(\text{MA60}, y_t^{(5)})$	0.27	0.38	0.42	0.44	0.46	0.44
(2)	(1.15)	(2.28)	(2.66)	(2.96)	(3.19)	(3.25)
(3) $\bar{R}^2$	0.07	0.14	0.17	0.19	0.21	0.20
(4) pval Wald (MA60, $y_t^{(5)}$)	0.71	0.49	0.48	0.47	0.50	0.56

Table A.4: Distribution of predictive regression statistics under random trends

The table shows distribution of the regression statistics obtained using random trends, tr_t^{RW} , generated as driftless random walks in 100 000 independent simulations. The regressions correspond to equations (24) and (25) in the paper, respectively, where in each we replace τ_t^{CPI} with the randomly generated trend tr_t^{RW} . Columns (1)–(7) summarize the mean, standard deviation, and quantiles of the distribution of the statistics from the simulations. For comparison, columns (8)–(9) report corresponding statistics from the data (same as Table 2 of the paper). Column (8) “Cycles/ τ_t^{CPI} ” reports the regressions that condition on τ_t^{CPI} ; column (9) uses $\bar{y}_t, y_t^{(1)}$ as the only regressors. T-statistics in predictive regressions in panel A are obtained using the reverse regression delta method. In panel B the t-statistic is Newey-West adjusted with 18 lags.

Simulated random trends							Data	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	Cycles/ τ_t^{CPI} (8)	Yields, $\bar{y}_t, y_t^{(1)}$ (9)
A. Single-stage regression, specification as in eqn. (24) in the paper								
$\bar{r}x_{t+1} = a + b_1 y_t^{(1)} + b_2 \bar{y}_t + b_3 tr_t^{RW} + \varepsilon_{t+1}$ (simulated)								
$\bar{r}x_{t+1} = a + b_1 y_t^{(1)} + b_2 \bar{y}_t + b_3 \tau_t^{CPI} + \varepsilon_{t+1}$ (data)								
	Mean	Std	P50	P10	P5	P1	P0.1	
t-stat of b_3	0.00	2.15	0.01	-2.88	-3.26	-3.84	-4.49	-4.65
$\bar{R}^2$	0.24	0.06	0.23	0.32	0.35	0.40	0.46	0.53
B. Cycles regression, specification as in eqn. (25) in the paper								
1st stage: $\bar{y}_t = \alpha + \beta tr_t^{RW} + \bar{c}_t^{RW}$ (simulated)								
$\bar{y}_t = \alpha + \beta \tau_t^{CPI} + \bar{c}_t$ (data)								
2nd stage: $\bar{r}x_{t+1} = a + b_1 c_t^{(1), RW} + b_2 \bar{c}_t^{RW} + \varepsilon_{t+1}$ (simulated)								
$\bar{r}x_{t+1} = a + b_1 c_t^{(1)} + b_2 \bar{c}_t + \varepsilon_{t+1}$ (data)								
	Mean	Std	P50	P90	P95	P99	P99.9	
1st stage t-stat of β	-0.01	4.40	-0.04	5.84	6.89	8.38	10.20	15.00
2nd stage $\bar{R}^2$	0.21	0.05	0.19	0.28	0.30	0.36	0.42	0.53

Table A.5: Forecasting nominal short rate

The table shows out-of-sample forecasts of the one-year nominal yield. The forecast horizon is from 12 through 60 months ahead. Column (1) reports the RMSE in basis points for the forecast using τ^{CPI} according to specification 1 (baseline); columns (2)–(4) report relative RMSE for alternative models, i.e. RMSE baseline/RMSE alternative. Numbers lower than one indicate that the model based on τ^{CPI} delivers more precise forecasts. As alternative models, we consider one-year yield as random walk (column (2)); DMA of the short rate τ^{y1} (column (3)); and a factor model using yield PCs as factors, with the level factor following random walk as in Duffee (2009) (column (4)). In panels A and B, we consider two- and three-factor models, respectively, as discussed in the text of Section II of the Appendix.

	(1)	(2)	(3)	(4)
h	RMSE (bps)	Ratio	Ratio	Ratio
months	(τ^{CPI})	(RW)	(τ^{y1})	(PCA)
A. Two factors				
Start of out-of-sample period: 1978:01				
12	189.02	1.01	0.85	0.96
24	248.03	0.92	0.80	0.90
36	283.67	0.91	0.79	0.93
48	281.27	0.89	0.75	0.92
60	287.66	0.96	0.74	0.94
Start of out-of-sample period: 1985:01				
12	148.26	1.01	0.88	1.00
24	191.40	0.84	0.80	0.91
36	198.28	0.73	0.75	0.87
48	200.58	0.74	0.73	0.89
60	218.82	0.91	0.73	1.02
Start of out-of-sample period: 1995:01				
12	136.48	0.98	0.92	1.05
24	187.09	0.81	0.83	0.94
36	197.36	0.69	0.77	0.89
48	192.57	0.66	0.74	0.87
60	197.38	0.77	0.74	0.97

	(1)	(2)	(3)	(4)
h	RMSE (bps)	Ratio	Ratio	Ratio
months	(τ^{CPI})	(RW)	(τ^{y1})	(PCA)
B. Three factors				
Start of out-of-sample period: 1978:01				
12	193.36	1.03	0.85	0.99
24	250.50	0.93	0.81	0.91
36	279.01	0.90	0.79	0.92
48	278.64	0.88	0.76	0.92
60	282.64	0.94	0.74	0.94
Start of out-of-sample period: 1985:01				
12	143.41	0.98	0.86	0.97
24	188.85	0.82	0.80	0.89
36	198.36	0.73	0.76	0.86
48	200.46	0.74	0.74	0.89
60	218.70	0.91	0.75	1.01
Start of out-of-sample period: 1995:01				
12	136.12	0.98	0.92	1.05
24	187.61	0.81	0.84	0.94
36	200.68	0.70	0.79	0.89
48	193.45	0.66	0.75	0.86
60	196.31	0.76	0.73	0.95

Table A.6: Bond excess returns: summary statistics

Panel A reports summary statistics for annual excess returns for US nominal Treasury bonds. Panel B reports the mean and standard deviation of duration-standardized excess returns. AR(1) denotes the first order autocorrelation coefficient (monthly frequency, annual holding period). Panel C reports the correlation between excess returns of different maturities. The numbers are in percentages. The data has monthly frequency, covering sample period 1971:11–2011:12.

Panel A. Bond excess returns						
	$rx^{(2)}$	$rx^{(5)}$	$rx^{(7)}$	$rx^{(10)}$	$rx^{(15)}$	$rx^{(20)}$
Mean	0.70	1.78	2.21	2.24	2.87	3.39
Stdev	1.94	6.19	8.48	11.51	16.71	22.19
AR(1)	0.94	0.93	0.93	0.93	0.93	0.92
Panel B. Duration-standardized excess returns						
Mean	0.35	0.36	0.32	0.22	0.19	0.17
Stdev	0.97	1.24	1.21	1.15	1.11	1.11
Panel C. Correlation of excess returns						
$rx^{(2)}$	1.00	.	.	.	.	.
$rx^{(5)}$	0.95	1.00	.	.	.	.
$rx^{(7)}$	0.90	0.99	1.00	.	.	.
$rx^{(10)}$	0.86	0.96	0.99	1.00	.	.
$rx^{(15)}$	0.81	0.92	0.96	0.99	1.00	.
$rx^{(20)}$	0.76	0.86	0.91	0.93	0.97	1.00

Table A.7: Comparison of one-year holding period excess returns: CMT, GSW and FB data

The table reports β 's and R^2 's from regressions of excess returns constructed from CMT data on GSW (**panel A**) and FB (**panel B**) counterparts. We consider a monthly sample 1971:11–2011:12 with maturities from two to ten (five) years for GSW (FB) data. Excess returns are defined over a one-year holding period.

	$rx^{(2)}$	$rx^{(3)}$	$rx^{(4)}$	$rx^{(5)}$	$rx^{(6)}$	$rx^{(7)}$	$rx^{(8)}$	$rx^{(9)}$	$rx^{(10)}$
Panel A. Regressions of rx from CMT on GSW									
β	1.04	1.03	1.04	1.05	1.06	1.05	1.04	1.04	1.03
$\bar{R}^2$	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.98
Panel B. Regressions of rx from CMT on FB									
β	1.04	1.01	1.02	1.05	—	—	—	—	—
R^2	0.99	0.99	0.99	0.99	—	—	—	—	—

Table A.8: Predicting actual bond portfolio returns

The table reports the predictive regressions of actual bond portfolio excess returns from CRSP on $\widehat{cf}_t$ at different investment horizons, $rx_{t+h/12}$, $h = 1, 3, 6, 9, 12$ months. The column labels reflect the maturity bin for each bond portfolio: from below two years to above ten years. The sample is from 1972:01 (where return for the $> 120m$ portfolio become available) through 2011:12. We use log returns; returns are in excess of Tbill rates (1-month Tbill for $h < 3$, 3-month Tbill for $3 \leq h < 6$ and 6-month Tbill for $h \geq 6$). 3- and 6-month Tbill rates are obtained from H.15 Fed table, 1-month Tbill is from CRSP. T-bill rates are converted to a continuous basis. T-statistics in parentheses are obtained from Hodrick reverse regression delta method of Wei and Wright (2013). All variables are standardized.

	12m < rx < 24m	rx < 36m	rx < 48m	rx < 60m	rx < 120m	$rx \geq 120m$
Panel A. $h = 1$ month						
$\widehat{cf}_t$	0.17	0.18	0.19	0.19	0.21	0.23
	(4.50)	(4.57)	(4.99)	(5.24)	(5.38)	(6.29)
$\bar{R}^2$	0.03	0.03	0.03	0.04	0.04	0.05
Panel B. $h = 3$ months						
$\widehat{cf}_t$	0.30	0.31	0.33	0.34	0.36	0.39
	(4.04)	(4.21)	(4.53)	(4.77)	(4.87)	(5.56)
$\bar{R}^2$	0.09	0.10	0.11	0.11	0.13	0.15
Panel C. $h = 6$ months						
$\widehat{cf}_t$	0.44	0.46	0.48	0.49	0.51	0.55
	(3.96)	(4.16)	(4.51)	(4.72)	(4.85)	(5.36)
$\bar{R}^2$	0.19	0.21	0.23	0.24	0.26	0.30
Panel D. $h = 9$ months						
$\widehat{cf}_t$	0.54	0.56	0.58	0.59	0.62	0.66
	(3.93)	(4.19)	(4.56)	(4.78)	(4.94)	(5.36)
$\bar{R}^2$	0.29	0.31	0.34	0.35	0.39	0.44
Panel E. $h = 12$ months						
$\widehat{cf}_t$	0.62	0.64	0.66	0.67	0.70	0.74
	(4.05)	(4.40)	(4.79)	(5.03)	(5.15)	(5.44)
$\bar{R}^2$	0.38	0.41	0.43	0.44	0.50	0.55

Table A.9: Comparing predictive R^2 in different data sets

The table compares the predictive $\bar{R}^2$'s for three different zero curves obtained from: Fama-Bliss (FB), Gürkaynak, Sack, and Wright (2006, GSW), and Treasury constant maturity (CMT) rates. The dependent variable is: $\bar{r}x_{t+1} = \frac{1}{4} \sum_{i=2}^5 rx_{t+1}^{(i)}$, and regressors are indicated in the first column. In both panels, row (1) and (1*) uses two cycles with maturity one and five years, row (2) and (2*): five cycles with maturities from one through five years; rows (1) and (2) construct the discounted moving average τ_t^{CPI} with core CPI inflation, and rows (1*) and (2*) with all-items CPI inflation; row (3): two yields with maturity one and five years; row (4): five forward rates with maturity one through five years; row (5): spread between five- and one-year cycle; row (6): spread between five- and one-year yield. The column "sample" provides adjusted $\bar{R}^2$'s for each regression; the column "bootstrap" gives the 5%, 50% and 95% percentile values for the adjusted R^2 's obtained with the stationary block bootstrap following Li and Maddala (1997).

Panel A. Pre crisis: 1971–2006						
CMT data			GSW data		FB data	
Regressor	Sample	Bootstrap	Sample	Bootstrap	Sample	Bootstrap
<i>Core CPI</i>						
(1) $c^{(1)}, c^{(5)}$	0.53	[0.38, 0.52, 0.63]	0.53	[0.39, 0.52, 0.63]	0.51	[0.37, 0.50, 0.61]
(2) $c^{(1)}, \dots, c^{(5)}$	0.53	[0.40, 0.54, 0.64]	0.53	[0.40, 0.53, 0.64]	0.55	[0.41, 0.56, 0.66]
<i>All items CPI</i>						
(1*) $c^{(1)}, c^{(5)}$	0.49	[0.31, 0.48, 0.61]	0.48	[0.29, 0.46, 0.60]	0.47	[0.29, 0.46, 0.59]
(2*) $c^{(1)}, \dots, c^{(5)}$	0.50	[0.33, 0.50, 0.62]	0.48	[0.31, 0.49, 0.61]	0.51	[0.32, 0.50, 0.64]
<i>Yields-only</i>						
(3) $y^{(1)}, y^{(5)}$	0.22	[0.10, 0.24, 0.39]	0.19	[0.08, 0.21, 0.36]	0.19	[0.08, 0.21, 0.37]
(4) $f^{(1)}, \dots, f^{(5)}$	0.27	[0.17, 0.30, 0.43]	0.21	[0.12, 0.25, 0.40]	0.30	[0.19, 0.32, 0.44]
<i>Spreads</i>						
(5) $c^{(5)} - c^{(1)}$	0.13	[0.02, 0.12, 0.23]	0.11	[0.02, 0.10, 0.22]	0.11	[0.02, 0.10, 0.22]
(6) $y^{(5)} - y^{(1)}$	0.13	[0.03, 0.12, 0.24]	0.11	[0.02, 0.11, 0.23]	0.11	[0.02, 0.11, 0.23]
Panel B. Post crisis: 1971–2011						
<i>Core CPI</i>						
(1) $c^{(1)}, c^{(5)}$	0.44	[0.25, 0.44, 0.57]	0.43	[0.24, 0.43, 0.57]	0.42	[0.24, 0.41, 0.55]
(2) $c^{(1)}, \dots, c^{(5)}$	0.46	[0.28, 0.47, 0.60]	0.45	[0.28, 0.46, 0.59]	0.45	[0.27, 0.46, 0.60]
<i>All items CPI</i>						
(1*) $c^{(1)}, c^{(5)}$	0.34	[0.13, 0.34, 0.53]	0.32	[0.11, 0.32, 0.52]	0.31	[0.12, 0.31, 0.50]
(2*) $c^{(1)}, \dots, c^{(5)}$	0.36	[0.17, 0.37, 0.55]	0.33	[0.15, 0.34, 0.53]	0.34	[0.15, 0.35, 0.55]
<i>Yields-only</i>						
(3) $y^{(1)}, y^{(5)}$	0.16	[0.05, 0.17, 0.32]	0.13	[0.04, 0.15, 0.30]	0.13	[0.04, 0.15, 0.29]
(4) $f^{(1)}, \dots, f^{(5)}$	0.17	[0.07, 0.21, 0.36]	0.14	[0.07, 0.18, 0.33]	0.20	[0.09, 0.23, 0.37]
<i>Spreads</i>						
(5) $c^{(5)} - c^{(1)}$	0.11	[0.02, 0.10, 0.21]	0.10	[0.01, 0.09, 0.20]	0.10	[0.01, 0.09, 0.20]
(6) $y^{(5)} - y^{(1)}$	0.12	[0.02, 0.11, 0.24]	0.10	[0.01, 0.10, 0.22]	0.10	[0.01, 0.10, 0.22]

Table A.10: Comparing alternative constructions of the cycle factor

The table reports correlations between alternative approaches to constructing the cycle factor. $\widehat{cf}_t$ is used in the body of the paper, and defined in equation (27); $\widehat{cf}_t^{CS}$ differs from the baseline specification in that it involves no information about future returns as discussed in Section IV.A; $\widehat{cf}_t^{PC}$ is obtained from the eigenvalue decomposition of expected excess returns in Section IV.C of this Appendix; $\widehat{cf}_t^{NLS}$ is obtained in a one-step estimation in Section IV.B.

	Correlations			
	$\widehat{cf}_t$	$\widehat{cf}_t^{CS}$	$\widehat{cf}_t^{PC}$	$\widehat{cf}_t^{NLS}$
$\widehat{cf}_t$	1	0.999	0.998	0.981
$\widehat{cf}_t^{CS}$	·	1	0.999	0.982
$\widehat{cf}_t^{PC}$	·	·	1	0.988
$\widehat{cf}_t^{NLS}$	·	·	·	1

Table A.11: Correlation of $\widehat{cf}$ with survey-implied expected returns

The table shows the unconditional correlation of survey-implied risk premia with the $\widehat{cf}^{RT}$ factor. We consider a range of available interest rate surveys: BCEI, BCFF and SPF, and obtain expected returns for two- and ten-year maturity. To be consistent with the surveys, $\widehat{cf}^{RT}$ is constructed in real time using zero yields from two data sets: Fama-Bliss and CMT. Since the BCEI survey is available semiannually, this is the frequency at which we compute the correlations. Due to data availability, the starting dates and the sample length used to compute the correlations differ across surveys. P-values are reported in parentheses.

	BCFF 2Y	SPF 2Y	BCEI 10Y	BCFF 10Y	$\widehat{cf}^{RT}$ CMT	$\widehat{cf}^{RT}$ FB
BCFF 2Y	1.00	0.74	0.56	0.75	0.62	0.76
pval		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
SPF 2Y		1.00	0.65	0.42	0.50	0.63
pval			(0.00)	(0.00)	(0.00)	(0.00)
BCEI 10Y			1.00	0.36	0.48	0.57
pval				(0.01)	(0.00)	(0.00)
BCFF 10Y				1.00	0.34	0.42
pval					(0.02)	(0.00)
$\widehat{cf}^{RT}$ CMT					1.00	0.84
pval						(0.00)
$\widehat{cf}^{RT}$ FB						1.00
pval						(0.00)

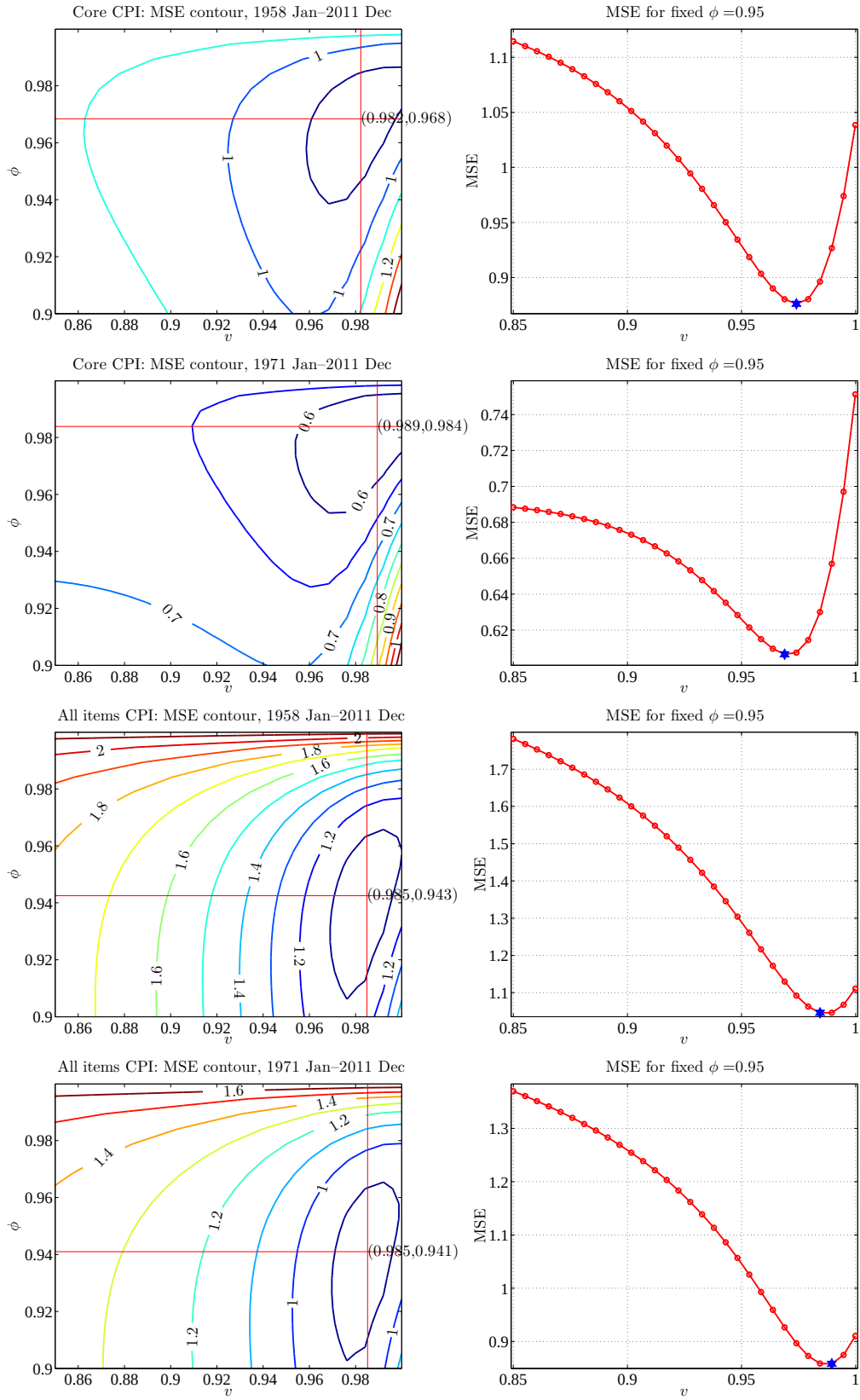


Figure A.1: Inflation MSEs for different gain parameters v

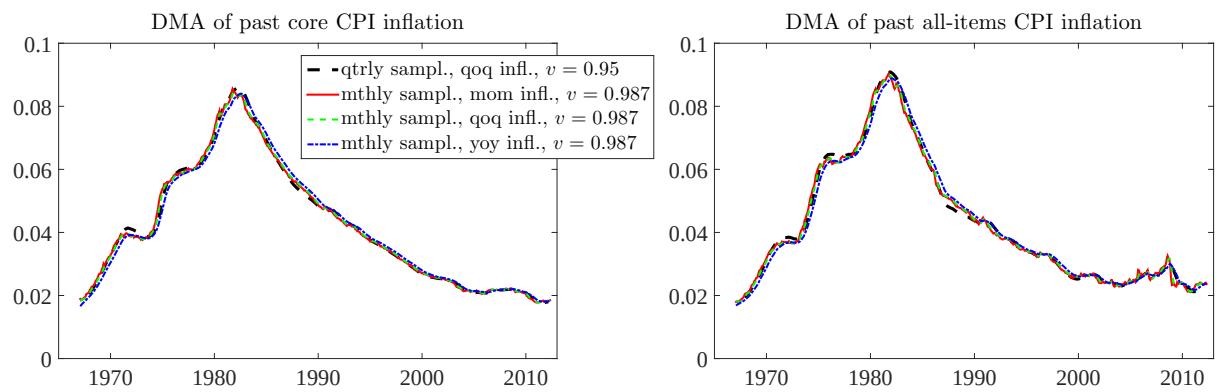


Figure A.2: Discounted moving average for different sampling frequencies and inflation horizons

The figure plots the discounted moving average (DMA) constructed for year-over-year, quarter-over-quarter and month-over-month inflation sampled monthly, as well as quarter-over-quarter inflation sampled quarterly. All inflation numbers are annualized. For monthly sampling the gain parameter is 0.987 and for quarterly sampling it is 0.95. The window length covers 10 years of data.

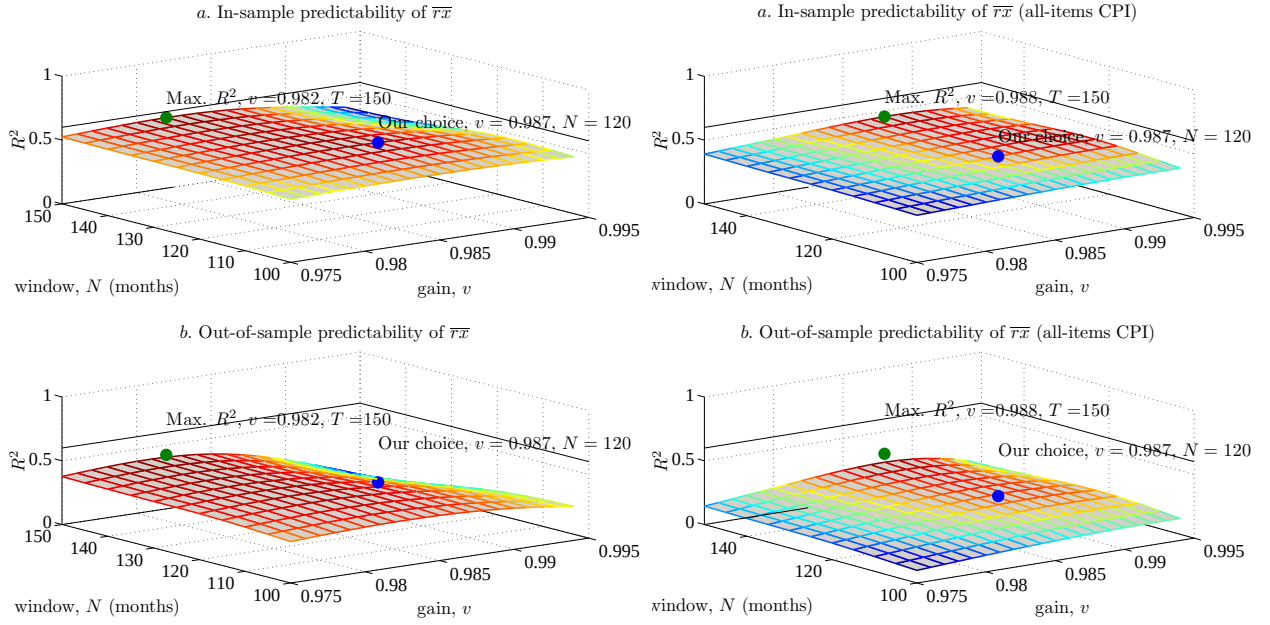


Figure A.3: Sensitivity of the predictive results to v and N parameters in constructing τ_t^{CPI}

The figure studies the sensitivities of predictive $\bar{R}^2$'s to the values of N and v parameters used to construct τ_t^{CPI} (equation (19) in the paper). The first column reports the results for τ_t^{CPI} obtained from the core CPI and the second column reports the results using all-items CPI. We predict the average bond return across maturities corresponding to specification (25) in the paper. We consider the gain parameter v between 0.975 and 0.995, and the window size between 100 and 150 months. Panel *a* and *b* in each column correspond to in-sample and out-of-sample forecasts, respectively.

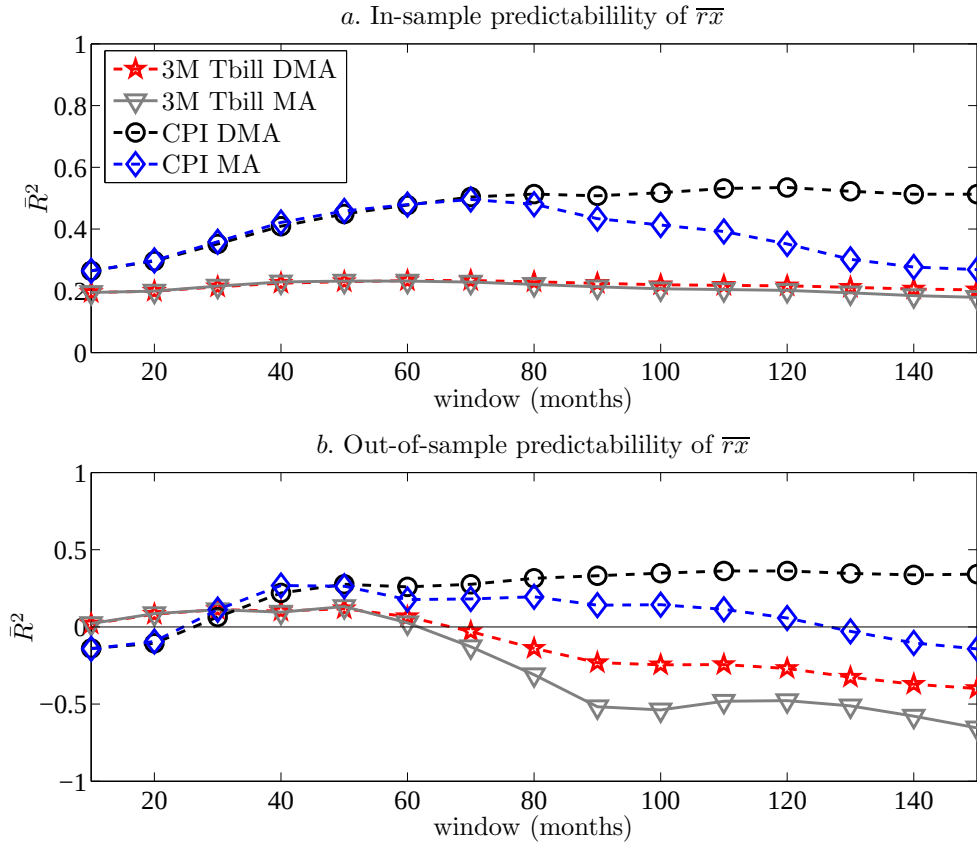


Figure A.4: Smoothing inflation vs Tbill rate

The figure depicts the sensitivity of the predictive $\bar{R}^2$'s, in- and out-of-sample, panels *a* and *b*, respectively, to the length of the moving average (MA) and discounted moving average (DMA) window. The window varies from ten to 150 months. We consider predictability of $\bar{r}\bar{x}_{t+1}$, and use moving average of past core CPI, and past three-month T-bill rate. The regressions are specified analogously to (3). The out-of-sample period in panel *b* starts in 1978:01. The sample is 1971:11-2011:12.

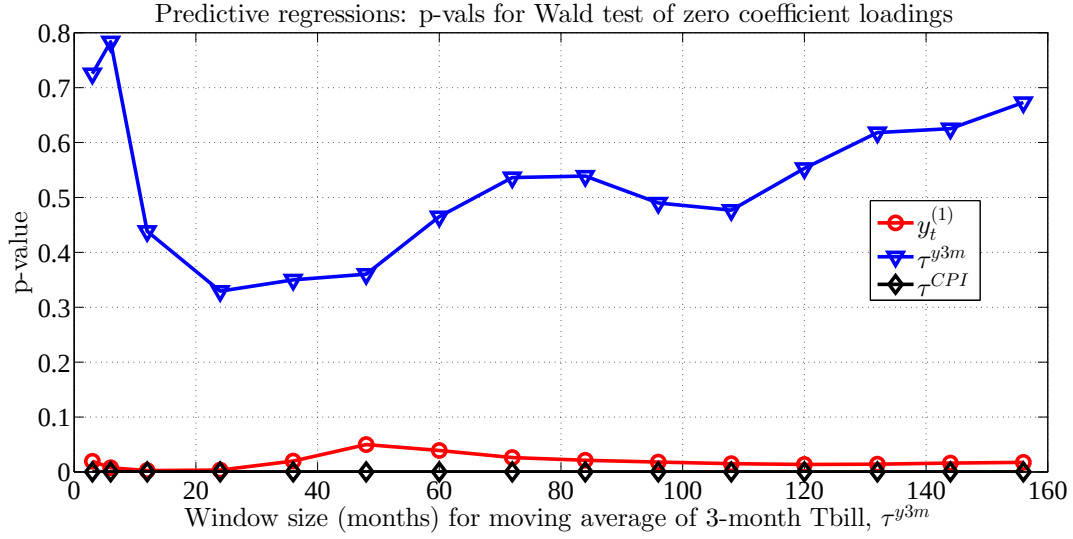


Figure A.5: Significance of the smoothed Tbill rate in predictive regressions

The figure reports the p-values of the restriction that a given coefficient in regression (4) is zero. We report the p-values for the zero restriction on the one-year yield $y_t^{(1)}$, inflation proxy τ_t^{CPI} , and the smoothed Tbill rate τ_t^{y3m} . Regression (4) is estimated for different levels of smoothing of the three-month Tbill rate considering window sizes from three months to 156 months (13 years): The window size for smoothing the Tbill rate is the only parameter that changes across regressions. τ_t^{CPI} is the discounted moving average of past core CPI using the same parameters as in the body of the paper. The reported numbers are average p-values across maturities.

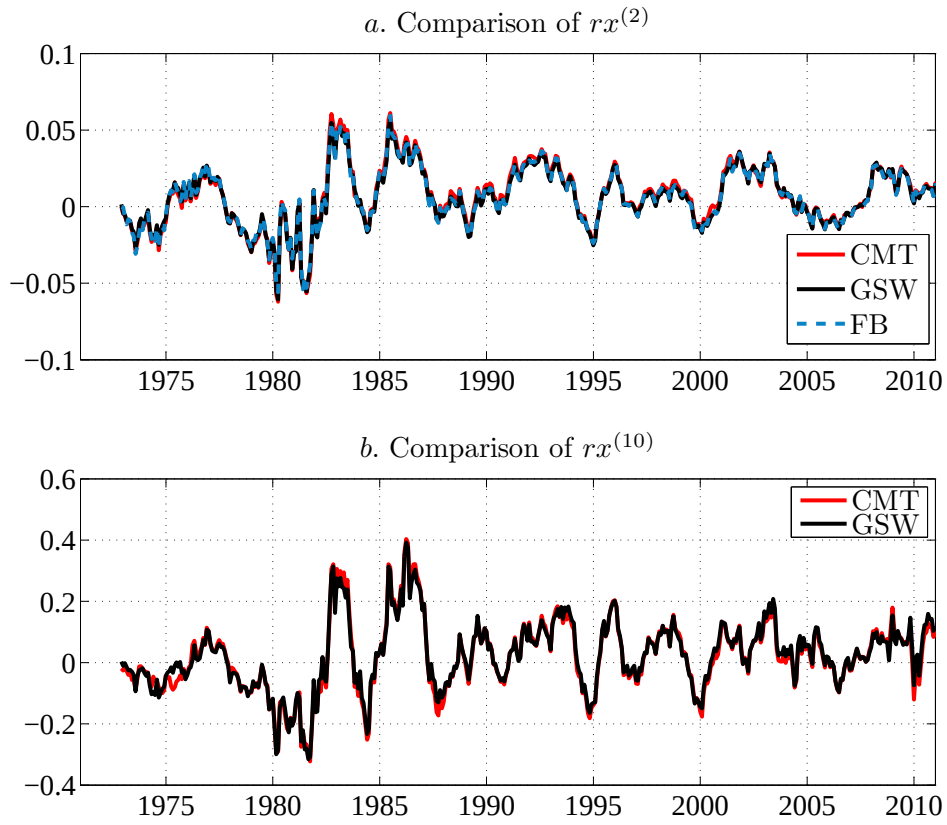


Figure A.6: Comparison of realized excess returns across data sets

The figure plots one-year holding period returns on zero bonds constructed from three data sets: CMT, GSW and FB over the period 1971:11–2011:12. Upper panel provides a comparison for the excess returns on a two-year bond, the bottom panel compares the excess returns on the ten-year bond.